

JEE MAIN REVISION MASTER: LIMITS, CONTINUITY AND DIFFERENTIABILITY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 271 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Limits Continuity And Differentiability - Core Application	271 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Limits Continuity And Differentiability - Core Application (Very High Yield — 271 PYQs)

Core Principle: Key problem pattern in Limits Continuity And Differentiability.

■■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

- Representative Source PYQs:
- [2026 Evening Shift] Q#1 (MCQ): "For the function $f(x)=e^{\sin|x|}-|x|, x \in R$, consider the following statements : Statement I : f is differentiable for all $x \in R$ " [Ans: N/A]
 - [2026 Morning Shift] Q#1 (MCQ): "The product of all possible values of α , for which $\lim_{x \rightarrow 0} \frac{1-\cos(\alpha x)\cos((\alpha+1)x)\cos(\alpha x)}{x^2}$ exists and is non-zero is" [Ans: N/A]
 - [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be a differentiable function in the interval $(0,\infty)$ such that $y(1)=2$, and $\lim_{x \rightarrow \infty} \frac{y(x)}{x^2} = \frac{1}{2}$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Let $f(x) = x^3 + 8$; $x < 0$, $x^2 - 4$; $x \geq 0$, and $g(x) = (x-8)^{1/3}$; $x < 0$, $(x+4)^{1/2}$; $x \geq 0$. Then the number of points, where the function $g \circ f$ is discontinuous, is ____.

Correct Answer: 3

Solution Analysis: We need to find the points where the composite function $g \circ f$ is discontinuous. That means we first find $(g \circ f)(x) = g(f(x))$. So we must see, for each value of x , which branch of g will be used. 1. Given functions $f(x) = x^3 + 8$, $x < 0$, $x^2 - 4$, $x \geq 0$ and $g(x) = (x-8)^{1/3}$, $x < 0$, $(x+4)^{1/2}$, $x \geq 0$. Now, $(g \circ f)(x) = g(f(x))$. The definition of g depends on whether its input is negative or non-negative. So we must check the sign of $f(x)$. 2. Find where $f(x) < 0$ and where $f(x) \geq 0$
Case 1: $x < 0$